



COLORADO DIVISION OF MINERALS AND GEOLOGY
MINERALS PROGRAM INSPECTION REPORT
PHONE: (303) 866-3567

The Division of Minerals and Geology has conducted an inspection of the mining operation noted below. This report documents observations concerning compliance of the mining operation with the permit and the regulations of the Mined Land Reclamation Board. The report notes 1) Areas of successful compliance; 2) Problems and suggested corrective actions and/or 3) Possible violations to be considered for possible enforcement action by the Mined Land Reclamation Board. **OPERATORS SHOULD READ THIS REPORT CAREFULLY BECAUSE IT MAY REQUIRE CORRECTIVE ACTION AND/OR RESPONSES TO THE DIVISION IN ORDER TO AVOID CONSIDERATION OF POSSIBLE ENFORCEMENT ACTION BY THE MINED LAND RECLAMATION BOARD.**

MINE NAME: Larimer Quarry OPERATOR: CEMEX, Inc.

COUNTY: Larimer MINERAL: SDG TYPE OF OPERATION: 112

INSPECTOR(S): Erica Crosby

MINE ID # OR PROSPECTING ID #: M-1977-341

INSPECTION DATE:	<u>12/06/02</u>	DATE OF COMPLAINT:	<u>NA</u>
INSPECTOR'S INITIALS:	<u>ESC</u>	TIME OF DAY (MILITARY):	<u>10:00</u>
INSPECTION TYPE CODE ⁽¹⁾ :	<u>SI</u>	POST INSP. CONTACTS ⁽²⁾ :	<u>NO</u>
JOINT INSP. AGENCY CODE ⁽²⁾ :	<u>CT</u>	REASON FOR INSP. CODE ⁽³⁾ :	<u>IE</u>
WEATHER CODE ⁽⁴⁾ :	<u>CR</u>	BOND CALCULATION TYPE ⁽⁵⁾ :	<u>NN</u>

OP. REP. PRESENT: Erica Crosby, Loretta Pineda & Deb Bell; DMG; George Martin; CEMEX, Inc. Terry Parrish; Landowner

1. **INSPECTION TYPE CODE** - [CL-IN: IL=Illegal Operation, MI=Monitoring, MP=Mineral Prospect, SI=Surety-related, PR=Pre-operation]
2. **POST INSPECTION CONTACTS AND JOINT INSPECTION AGENCY CODE** - [CL-AG: NO=None, BL=BLM, CH=Colo. Dept. Health; CL=Land Board, CT=Citizen; CW=Wildlife, FS=Forest Service, HW=Hwy. Dept., LG=Local Government, SE=State Engr.]
3. **REASON FOR INSPECTION CODE** - [CL-RS: AG=Other Agency Request, CT=Citizen Complaint, IE=Normal I&E Program, HP=High Priority, PY=Priority]
4. **WEATHER CODE** - [CL-WE: CL=Cloudy, CR=Clear, IN=Inclement - prevented inspection, RN=Raining, SN=Snowing, WD=Windy]
5. **BOND CALCULATION TYPE** - [BC=Complete Bond, BP=Partial Bond, NN=None]

This list identifies the environmental and permit parameters inspected and gives a categorical evaluation of each. **IF PB OR PV IS INDICATED, YOU SHOULD READ THE FOLLOWING PAGES CAREFULLY IN ORDER TO ASSURE COMPLIANCE WITH THE TERMS OF YOUR PERMIT AND APPLICABLE RULES AND REGULATIONS.** If PV is indicated, you will be notified under separate cover when the Mined Land reclamation Board will consider possible enforcement action.

GENERAL INSPECTION TOPICS

(AR) RECORDS.....	<u>Y</u>	(FN) FINANCIAL WARRANTY.....	<u>Y</u>	(RD) ROADS.....	<u>Y</u>
(HB) HYDROLOGIC BALANCE.....	<u>Y</u>	(BG) BACKFILL & GRADING.....	<u>Y</u>	(EX) EXPLOSIVES.....	<u>Y</u>
(PW) PROCESSING WASTE/TAILING.....	<u>Y</u>	(SF) PROCESSING FACILITIES.....	<u>Y</u>	(TS) TOPSOIL.....	<u>Y</u>
(MP) GENL MINE PLAN COMPLIANCE.....	<u>Y</u>	(FW) FISH & WILDLIFE.....	<u>Y</u>	(RV) REVEGETATION.....	<u>Y</u>
(SM) SIGNS AND MARKERS.....	<u>Y</u>	(SP) STORM WATER MGT PLAN.....	<u>Y</u>	(SB) COMPLETE INSP.....	<u>Y</u>
(ES) OVERBURDEN/DEV. WASTE.....	<u>Y</u>	(SC) EROSION/SEDIMENTATION.....	<u>Y</u>	(RS) RECL PLAN/COMP... ..	<u>Y</u>
(AT) ACID OR TOXIC MATERIALS.....	<u>NA</u>	(OD) OFF-SITE DAMAGE.....	<u>NA</u>	(ST) STIPULATIONS.....	<u>NA</u>

Y = Inspected and Found in Compliance
N = Not Inspected

PV = Inspected and Possible Violations Noted
NA = Not Applicable

PB = Inspected and Problems Noted

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OBSERVATIONS

This was an inspection of the Larimer Quarry conducted by Erica Crosby of the Division of Minerals and Geology. Loretta Pineda and Deb Bell of DMG were present to assist staff using the permit imaging system in the field. George Martin of CEMEX, Inc. and Terry Parrish, the landowner, were present during the inspection.

CEMEX, Inc. filed a total release request with the Division on October 17, 2002. The purpose of this inspection was to evaluate the site in terms of the reclamation standards noted in the approved permit and the Construction Materials Rules and Regulations.

The Larimer Quarry is a surface limestone operation with a permit area of 208 acres. The permit was issued on October 26, 1978 to Marten Marietta Cement for a 196-acre area. The permit was amended in 1987 to add an additional 12.8 acres to the south. On August 17, 2001, the permit was transferred to CEMEX, Inc.

The mine plan states that the limestone was hauled via truck 10-miles to the south to the Lyons operation, Permit No. M-1977-208. The quarry is divided into 4 distinct mining areas, Pits A, B, C and D. Approximately 250,000 tons of limestone has been mined from the site. Mining and reclamation occurred concurrently until the mid-1990's.

The reclamation plan calls for slopes to be reclaimed to 3h:1v and reference is also made to 2h:1v reclaimed slopes. The required topsoil replacement depth ranges from 6"-10". The reclamation plan calls for the site to be seeded with a dryland seed mixture at a rate of 25 to 75 pounds per acre. Ammonium nitrate fertilizer is then applied at a rate of 200 to 300 pounds per acre to facilitate growth.

The Division has specific comments regarding the site;

Backfilling/Grading & Topsoil Replacement- All pits have been backfilled, graded and topsoiled. On average, pit slopes are 3h:1v or less. On portions of the B and D pits, slopes appear to be steeper but remain less than 2h:1v as required by the permit. The A pit shows signs of subsidence along the southern extent of the pit. The subsidence crack varies from 1 foot to 2 feet in horizontal displacement. Vegetation has established around the feature, and does not hinder the approved postmining land use of rangeland.

The landowner expressed concerns regarding rocks that were pushed over the edge of the B Pit into undisturbed land. The rocks are located west of the pit, on the other side of the ridge where mining did not occur. This area, however, is located within the permit boundary. The rocks vary in size from 6" in diameter to over 2 feet in diameter. A review of the annual reports for this site indicates that mining in this quarry took place in the 1990's. The rocks are remnants of the blasted overburden and limestone mined from the quarry. The natural slope of the undisturbed area is 2h:1v or less. The Division does not believe that these rocks conflict with the postmining land use of rangeland. Access to these steep slopes is limited in nature, and the additional rocks should not pose any additional hazard or conflicts to the postmining land use of rangeland.

The C Pit is a south facing slope reclaimed to a gentle slope. At the base of the slope, water has concentrated in areas creating a number of rills and gullies. These features will need to be stabilized, repaired, and revegetated.

Revegetation- Portions of the B and D Pits on the steeper slopes, contain rocky soils with sparse vegetation. It appears that some of the topsoil has washed away, and vegetation has not established. The operator will need to test the soils to determine if any amendments need to be incorporated into the soil, and revegetate certain areas that have not established. The remaining slopes of the B and D Pits are well vegetated and no evidence of noxious weeds was noted.

In the lower portion of the C Pit a number of rills and gullies have formed where runoff concentrates. These areas will need to be stabilized and revegetated according to the reclamation plan. In addition, the west facing slope (appx. 3h:1v grade) on the eastern edge of the C Pit contains no perennial grasses. The only vegetation noted on this slope was Giant Evenstar (*Mentzelia decapitals*). The stand of plants covers approximately one

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acre of the slope. The operator will need to control the plant stand and reseed the slope in accordance with the approved seed mixture.

Existing Roads on Site- The mining and reclamation plan makes note that one road leading through the property will be relocated as a result of the mining operation. The relocated road is depicted on the Reclamation Plan Map dated June 11, 1987. Since this road is located on the reclamation plan map, and provides access to adjacent properties it can remain as a part of the postmining land use. However, a number of other roads are noted on site, but not included in the reclamation plan. These roads provide access to the A and C Pits. If the operator wishes to retain these roads as part of the postmining land use, the following items will need to be submitted:

1. A new Reclamation Plan Map and description of the road(s) will need to be submitted in the form of a revision to the permit.
2. The operator will need to obtain a letter from the landowner requesting the road(s) remain as a permanent feature.

Division Recommendation- Based on the outstanding issues noted above, the Division must deny the total bond release request as submitted by CEMEX, Inc. CEMEX may resubmit a partial release request (in the form of an acreage reduction) for areas that were deemed acceptable by the Division. The remaining areas and issues that were deemed not acceptable for the reasons noted above would remain in the permit until such items have been corrected. The bond in the amount of \$28,000.00 is adequate at this time.

Under Rule 4.18(1), any person directly and adversely affected or aggrieved by an Office decision to approve or deny the request for reclamation responsibility release, may appeal the decision to the Board by submitting a request for Administrative Appeal to the Office according to the provisions of Section 1.4.11

I & E Contact Address

NAME	<u>Mr. John Lohr</u>	☐ CE
OPERATOR	<u>CEMEX, Inc.</u>	☐ BL
STREET	<u>P.O. Box 529</u>	☐ FS
CITY/STATE/ZIP	<u>Lyons, CO 80540</u>	☐ HW
		☐ HMWMD (CH)
		☐ SE
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Cc: Carl Mount; DMG (via electronic transmission)

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